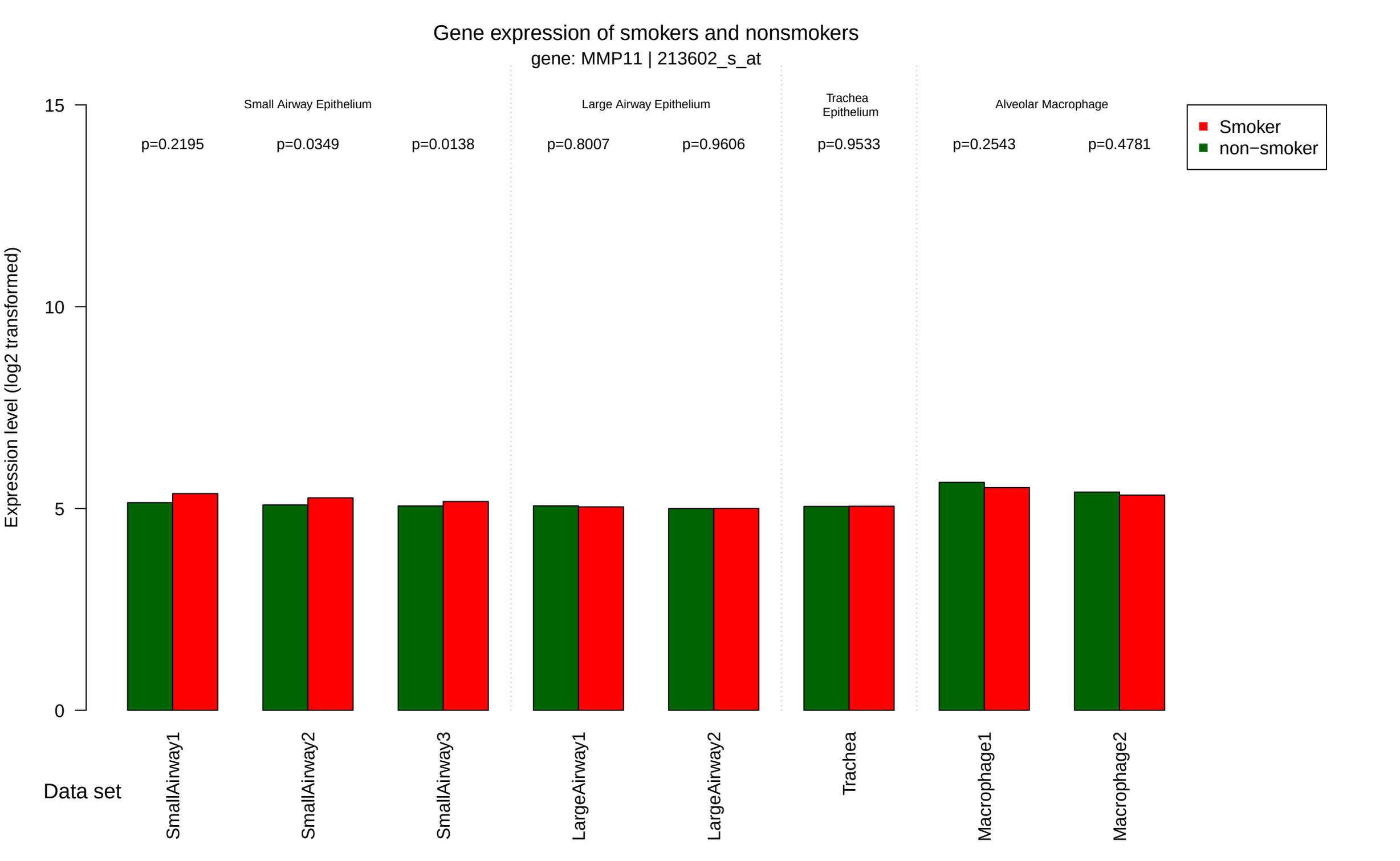
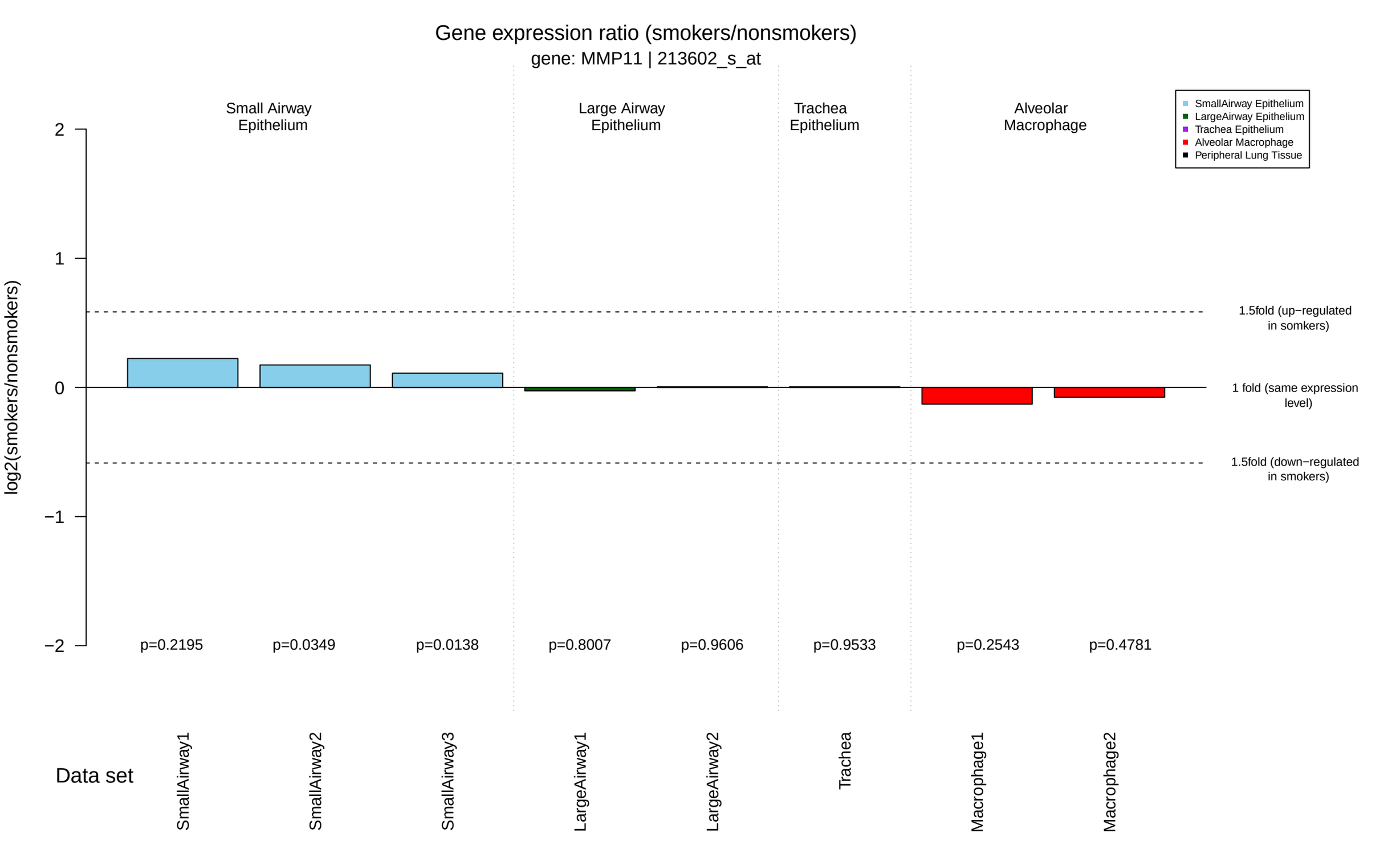
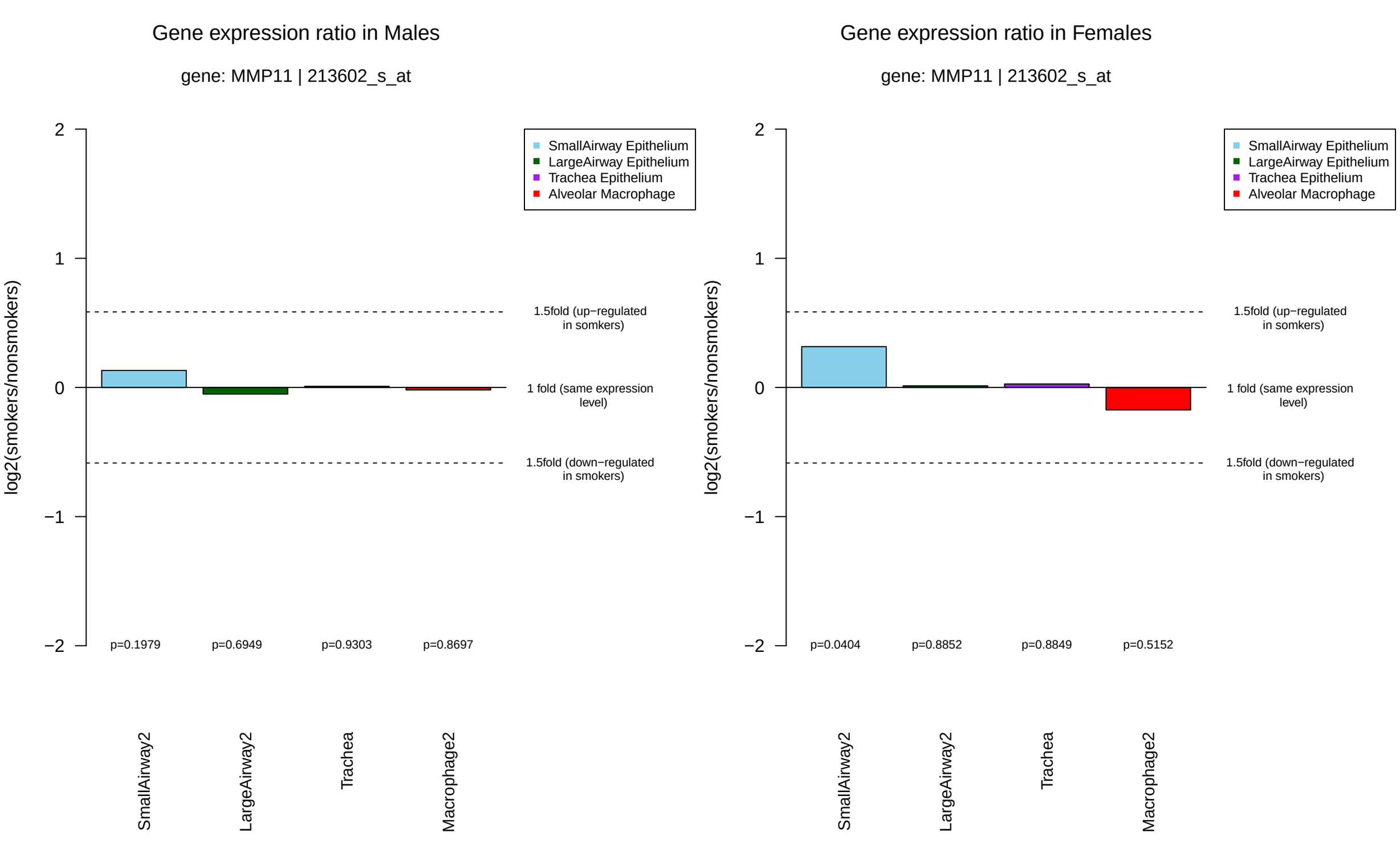
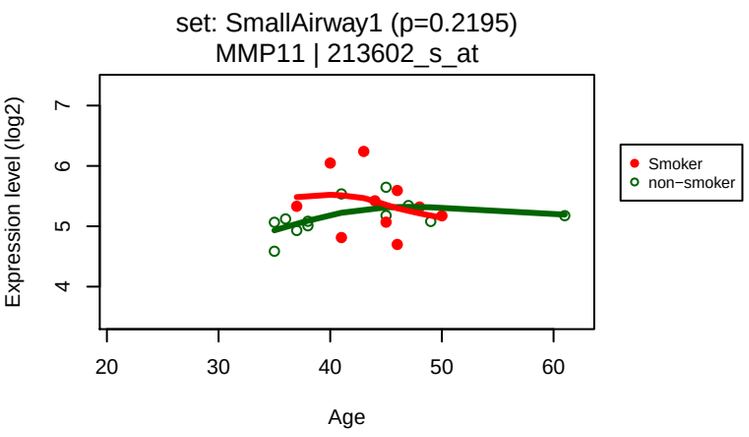
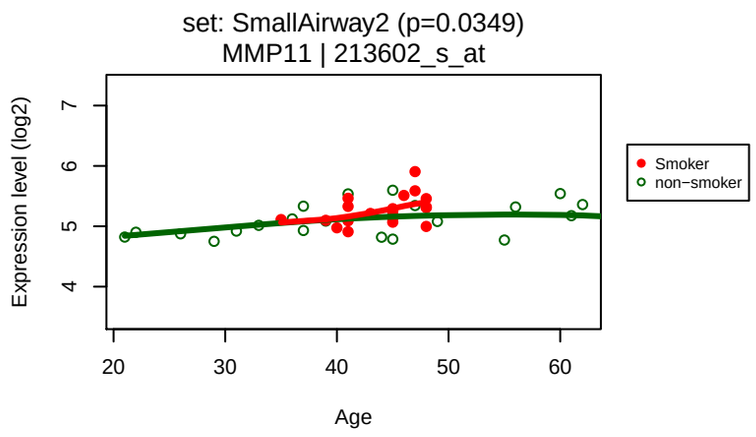
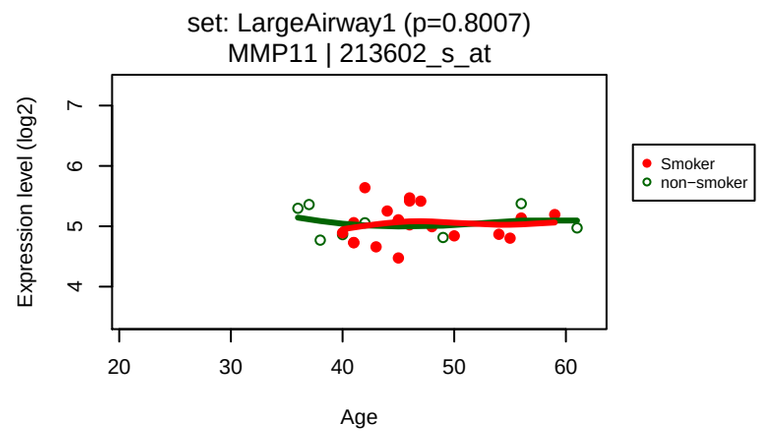

Gene expression of smokers and nonsmokers
gene: MMP11 | 213602_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2195
p=0.0349
p=0.0138
p=0.8007
p=0.9606
p=0.9533
p=0.2543
p=0.4781
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

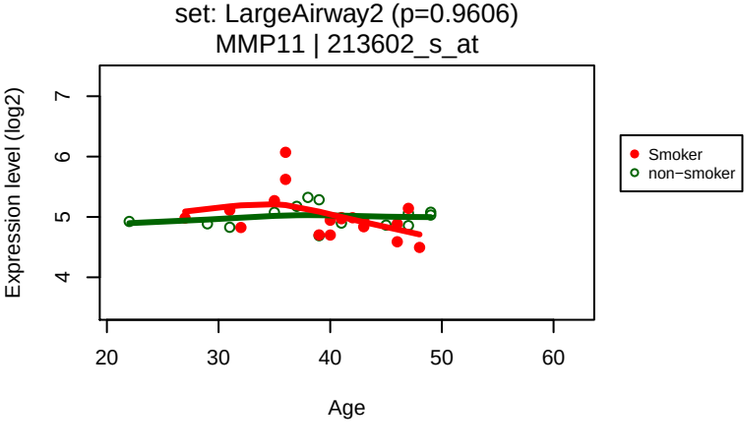

Gene expression ratio (smokers/nonsmokers)
gene: MMP11 | 213602_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2195
p=0.0349
p=0.0138
p=0.8007
p=0.9606
p=0.9533
p=0.2543
p=0.4781
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

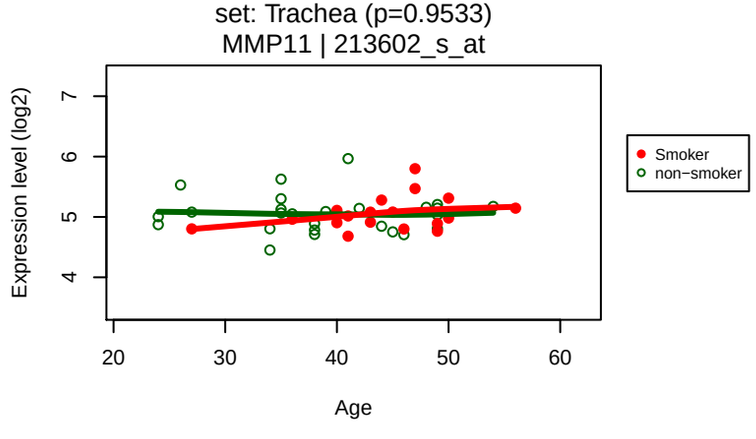

Gene expression ratio in Males
gene: MMP11 | 213602_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1979
p=0.6949
p=0.9303
p=0.8697
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP11 | 213602_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0404
p=0.8852
p=0.8849
p=0.5152
SmallAirway2
LargeAirway2
Trachea
Macrophage2

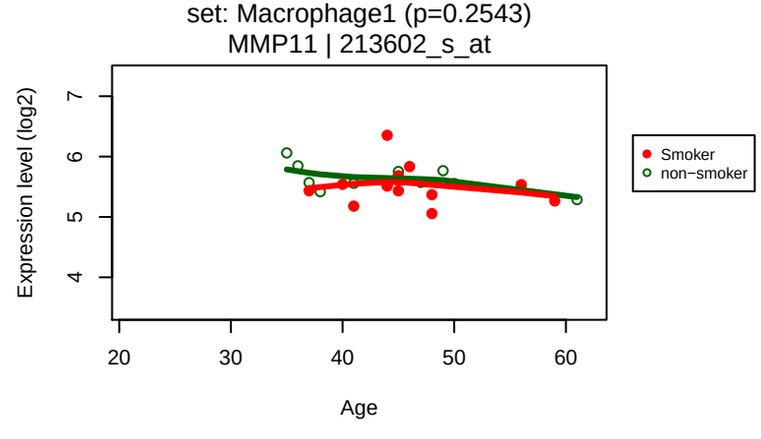

set: SmallAirway1 (p=0.2195)
MMP11 | 213602_s_at
Expression level (log2)
Age
Smoker
non−smoker

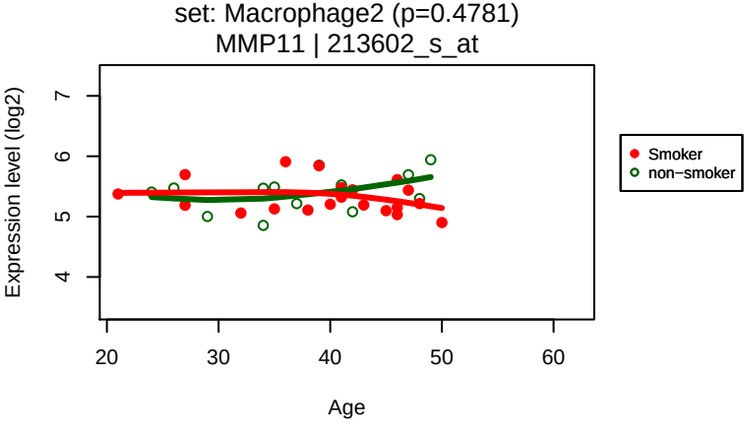

set: SmallAirway2 (p=0.0349)
MMP11 | 213602_s_at
Expression level (log2)
Age
Smoker
non−smoker

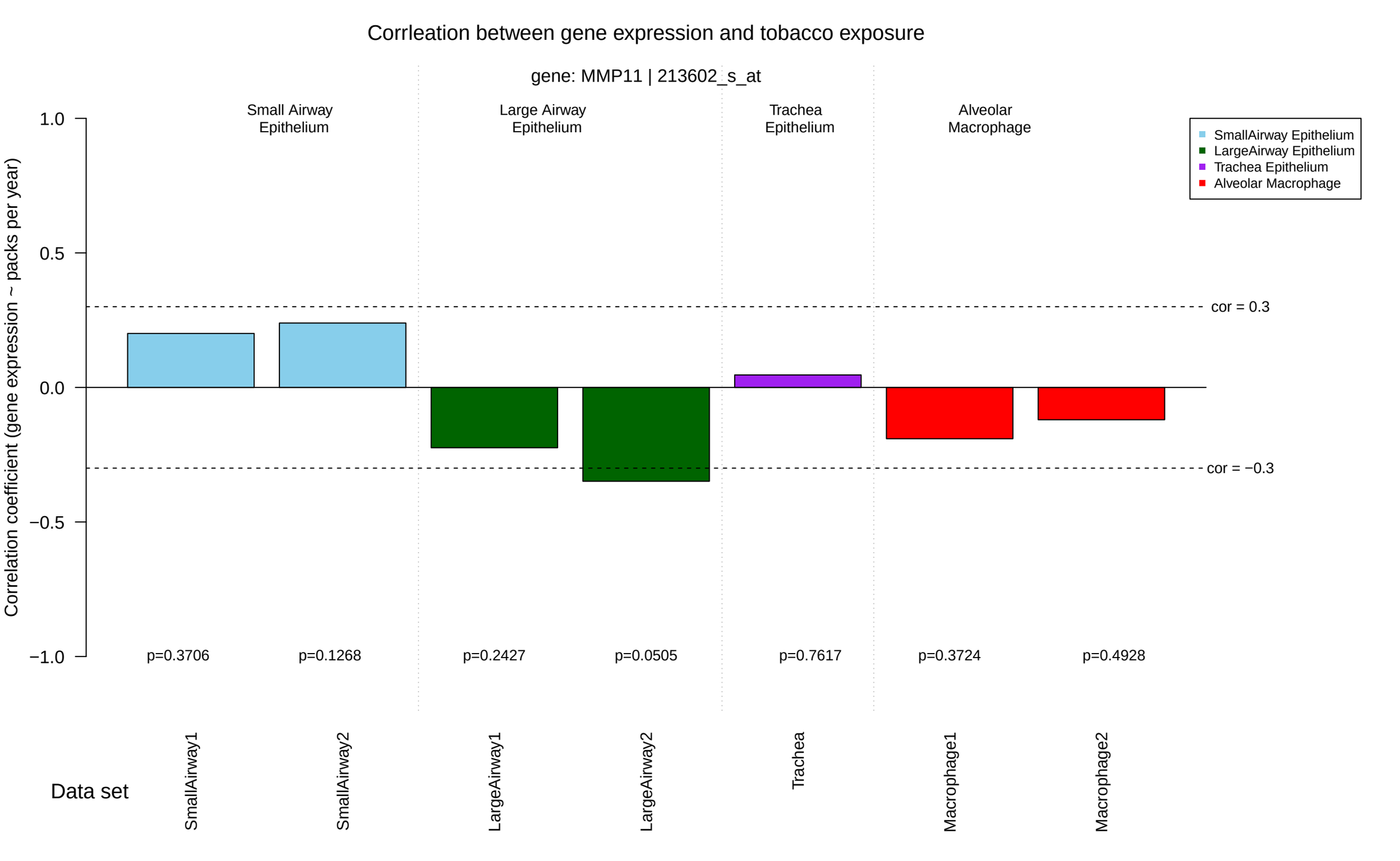

set: LargeAirway1 (p=0.8007)
MMP11 | 213602_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9606)
MMP11 | 213602_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9533)
MMP11 | 213602_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2543)
MMP11 | 213602_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4781)
MMP11 | 213602_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP11 | 213602_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3706
p=0.1268
p=0.2427
p=0.0505
p=0.7617
p=0.3724
p=0.4928
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2